

Treatment of Type 1 Reaction (T1R)

The treatment of Type 1 Reaction (T1R) primarily consists of corticosteroids, starting with **30–40 mg prednisone daily**, tapered gradually over a period of **4–12 months**. The tapering should be guided by clinical assessments such as **graded sensory testing** (using Semmes-Weinstein monofilaments) or **voluntary muscle testing**. During this period, the dose should remain at least **15–20 mg daily** to maintain effectiveness.

Adequate immunosuppression, defined as **>0.25 mg/kg/day**, should be maintained for at least **3 months in borderline tuberculoid (BT) leprosy**, and for **6–12 months or longer in borderline lepromatous (BL) leprosy**. In some cases, **dapsone** may help prevent recurrence of reactions.

Although the **WHO recommends only 3 months of steroid therapy**, this approach is not widely accepted by most clinicians, who favor longer treatment durations based on clinical response.

Treatment of Type 2 Reaction (T2R)

Type 2 Reaction (T2R), often presenting as Erythema Nodosum Leprosum (ENL), is more challenging to manage. The reaction is typically **episodic**, with **95% of ENL episodes lasting less than one month**.

- **Mild reactions** can be managed with **NSAIDs**.
- **Arthritis** associated with T2R may respond to **anti-malarial drugs**.

For **severe reactions**, the **WHO recommends**:

A standard course of **prednisolone** at a dose not exceeding **1 mg/kg/day**, with a total duration of **12 weeks**.

However, clinical experience suggests that **severe reactions often require higher doses (60–120 mg/day)** for a **short duration**, followed by rapid tapering to zero within **one month or less**. Recurrent attacks should be

treated similarly. This aggressive short-course approach is not yet universally accepted.

Attempts to **prevent recurrent reactions with prolonged steroid use** are limited by poor efficacy and significant side effects. Since T2R is an **immune complex-mediated reaction**, long-term steroid use often leads to **steroid dependence**, severe adverse effects, and in some cases, death.

Alternative and more effective options include:

- **Clofazimine**: Can prevent or treat T2R.
- **Thalidomide**: Superior to other treatments and effective both therapeutically and as prophylaxis. However, it may not work in all cases.

Combining low-dose steroids with low-dose thalidomide is counterproductive and should be avoided.

A major concern with **thalidomide** is its potential to cause **sensory neuropathy**, which can be difficult to distinguish from nerve damage due to leprosy itself.

When **thalidomide is unavailable**, **methotrexate (MTX)**—with or without additional clofazimine—can be considered for preventing recurrent ENL episodes.

Surgical Intervention

If **nerve function continues to deteriorate despite optimal medical therapy**, **nerve decompression (nerve release) surgery** should be considered.

Surgical evaluation may also be appropriate for **nerves that remain tender after treatment**, even in the absence of an active reaction.

Lucio's Phenomenon

Lucio's phenomenon is a rare, severe form of leprosy reaction seen exclusively in **untreated diffuse lepromatous leprosy**. It presents as **well-demarcated**,

bizarre-shaped superficial skin ulcers with slough (eschar) formation, resulting from **vasculitis and infarction** due to massive bacillary occlusion of small venules.

It may be triggered by **sudden exposure to cold**.

Treatment: Immediate initiation of **Multidrug Therapy (MDT)**, with **rifampicin (RMP)** being the most critical component.

Management of Leprosy Sequelae

The long-term consequences of leprosy—**sensory loss and muscle weakness**—are the main causes of disability and social stigma.

Management includes:

- Provision of **padded footwear, tools, and utensils** to prevent injury.
- **Tendon transfer surgery** may be considered to improve hand grip or foot function, but only under the guidance of an **experienced physiotherapist**.

Stigma reduction is essential and best achieved through **education**, emphasizing that **properly treated patients pose no risk and have minimal complications**.

Prevention

Despite ongoing research into **DNA-based vaccines**, **BCG vaccination** remains the most effective available prophylaxis in many regions. Protection varies widely—from **<20% to 80%**—likely due to differences in exposure to **environmental mycobacteria**.

The **WHO** recommends: - A **single 600 mg dose of rifampicin** for **contacts of leprosy patients**, which provides protection for several years. - **BCG vaccination** may extend this protective effect.

A newer WHO-endorsed strategy involves a longer initial prophylactic regimen using rifampicin in combination with moxifloxacin or clarithromycin for enhanced and prolonged protection.